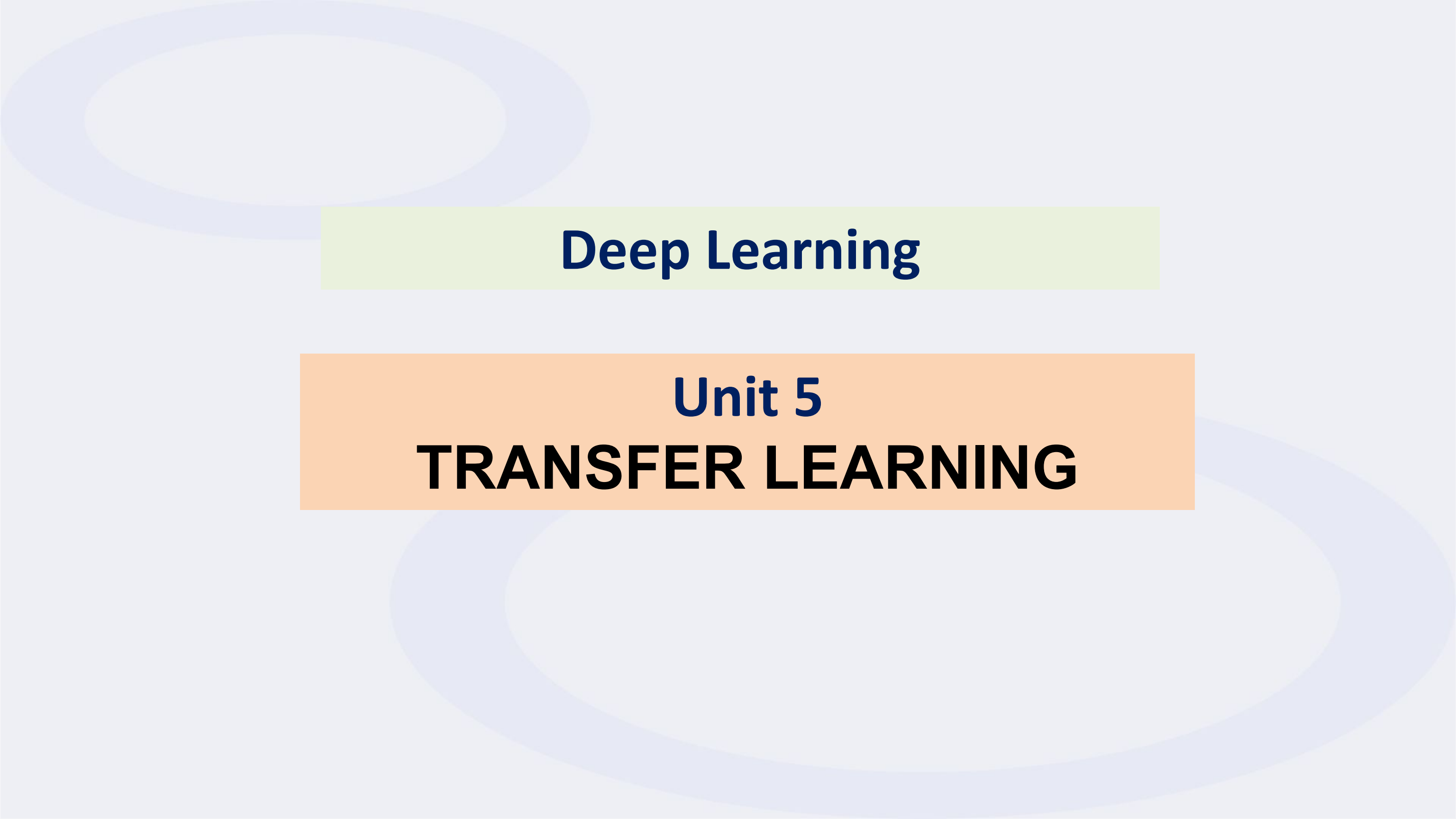




Deep Learning

Unit 5

TRANSFER LEARNING

Transfer Learning In Deep Learning

- Transfer learning is a technique in machine learning where a model trained on one task is reused as the starting point for a model on a second task.
- This approach can save time and resources, and can often lead to better performance than training a model from scratch.

Source Task

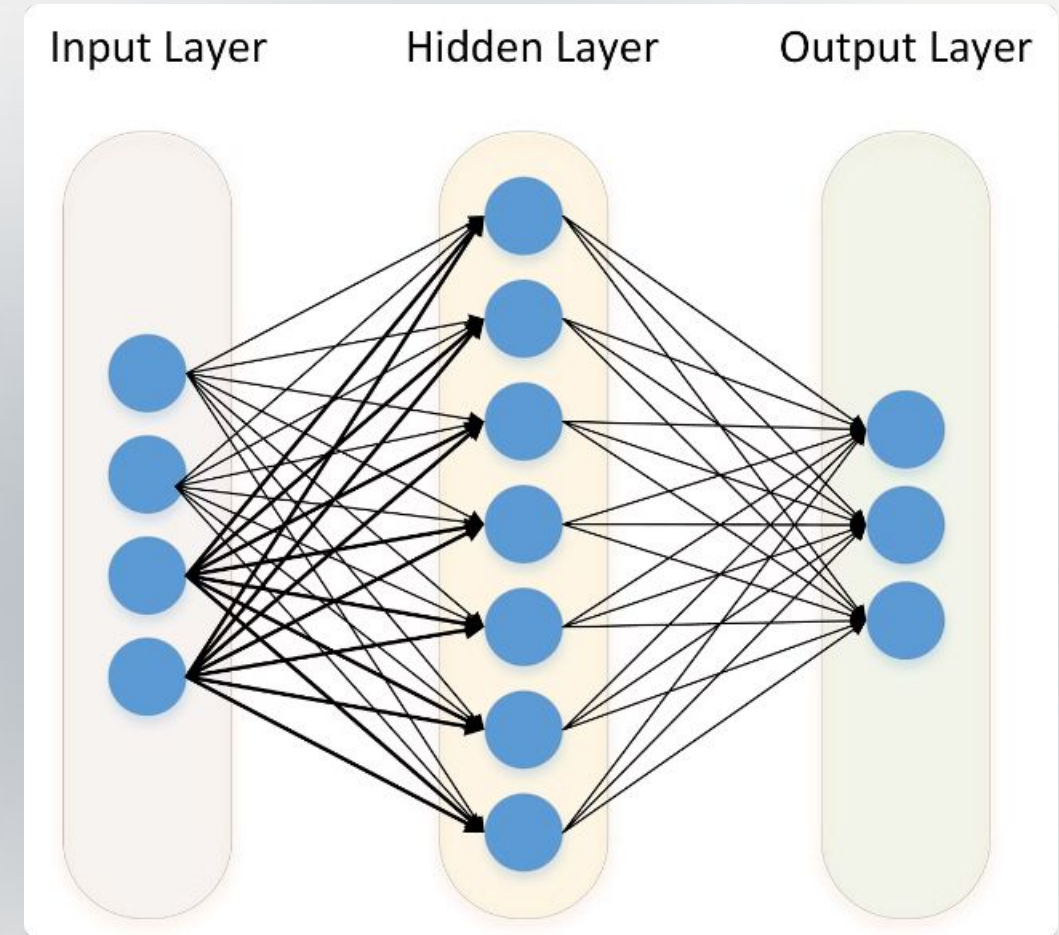
The original task the pre-trained model was trained on.

Target Task

The new task we want to adapt the model to.

Knowledge Transfer

The process of reusing knowledge from the source task to improve performance on the target task.



What is Transfer Learning and
Why Does it Matter?



Advantages of Transfer Learning

Transfer learning offers several advantages over training a model from scratch, such as improved accuracy, faster training times, and the ability to work with smaller datasets.

1

Improved Accuracy

Transfer learning can often achieve better performance than training a model from scratch, especially when working with limited data.

2

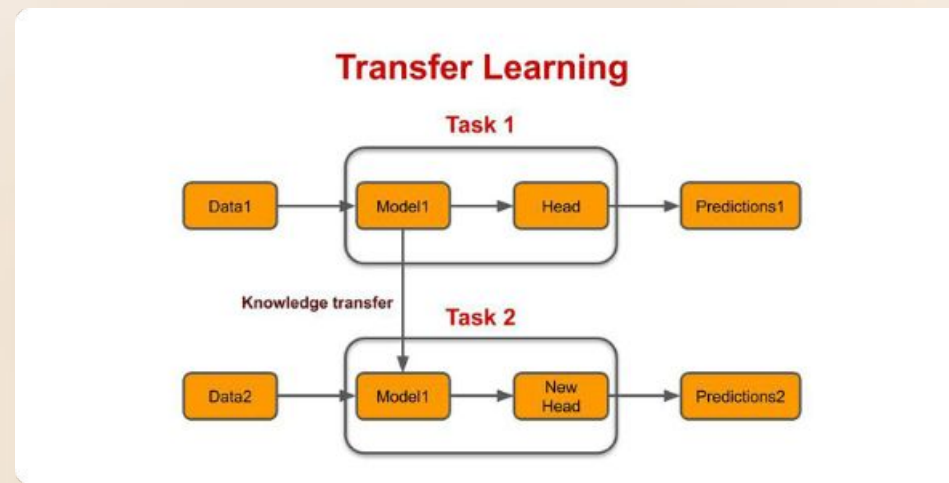
Faster Training Times

Transfer learning can significantly reduce training times by leveraging the knowledge gained from a pre-trained model.

3

Ability to Work with Smaller Datasets

Transfer learning can be used to train models on smaller datasets than would be required to train a model from scratch.



Concept of Transfer Learning

Transfer learning leverages the knowledge gained from solving one problem to improve the performance on a related but different problem. This approach aims to reduce the amount of data and time needed to train a new model.

Pre-trained Models

These models are trained on large datasets and have learned general features that can be applied to new tasks.

Fine-Tuning

The pre-trained model is adapted to the new task by adjusting its weights and biases.

Architecture of Transfer Learning

Transfer learning architecture consists of two main parts: a pre-trained model and a fine-tuning stage.

1

Pre-trained Model

The pre-trained model is trained on a large dataset and has learned general features that can be applied to new tasks.

2

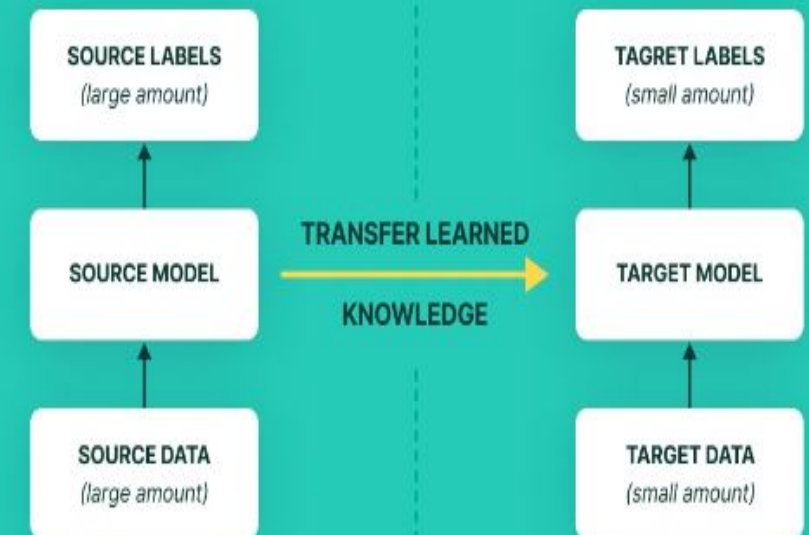
Fine-tuning

The pre-trained model is adapted to the new task by adjusting its weights and biases.

3

Target Task

The fine-tuned model is then used to perform the target task, such as image classification or object detection.



Types of Transfer Learning

There are three main types of transfer learning: inductive, transductive, and unsupervised.

Inductive Transfer Learning

The source and target domains have different feature spaces, but the tasks are similar.

Transductive Transfer Learning

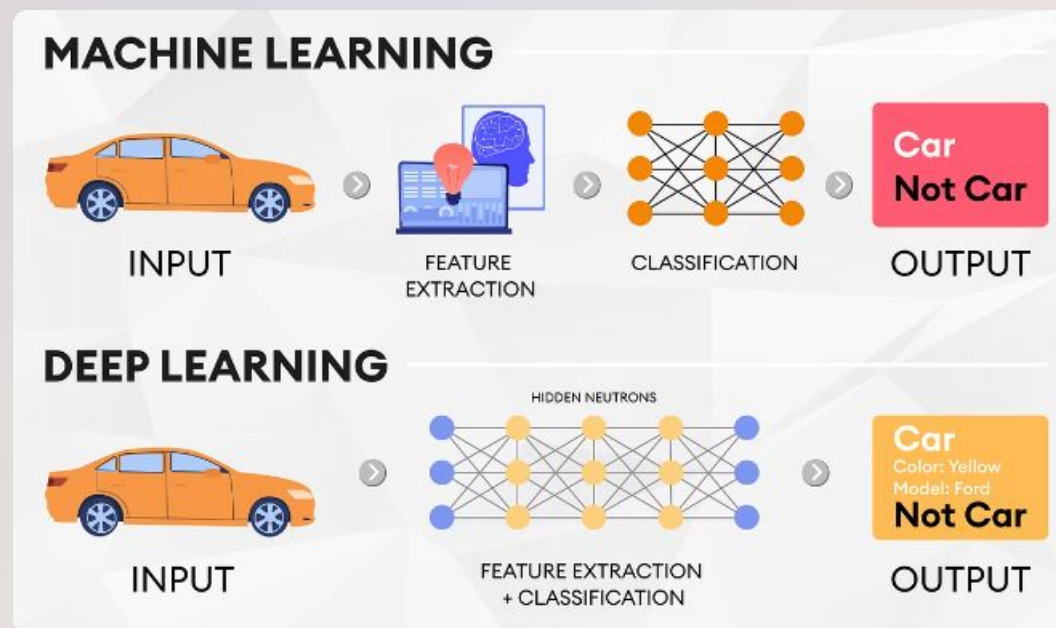
The source and target domains have the same feature space, but the tasks are different.

Unsupervised Transfer Learning

The source and target tasks are both unsupervised.

Challenges in Deep Learning

Deep learning models often require large amounts of data to train effectively. Obtaining and labeling this data can be expensive and time-consuming. Another challenge is that deep learning models can be difficult to interpret.



1

Data Quantity

Deep learning models can require large amounts of data to train effectively. Obtaining and labeling this data can be expensive and time-consuming.

2

Model Interpretability

Deep learning models can be difficult to interpret, making it challenging to understand why a model makes a particular prediction.

3

Computational Resources

Deep learning models often require significant computational resources to train and run, which can be a barrier for some researchers and developers.

Applications of Transfer Learning

Transfer learning has a wide range of applications, including image classification, object detection, natural language processing, and more.

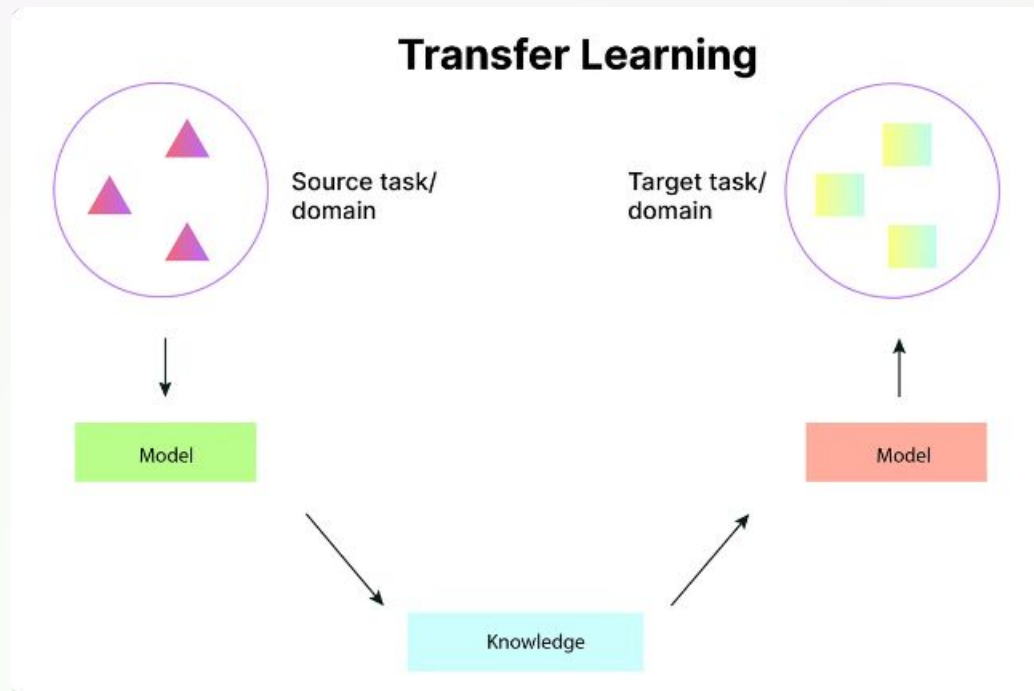


Image Classification

Object Detection

Natural Language
Processing

Image Recognition

Image
Segmentation

Machine
Translation



Conclusion and Future Direction

Transfer learning is a powerful technique for improving the performance of deep learning models. It's a rapidly evolving field with promising potential for further development and applications.



Advancements in Model Architecture

New architectures are being developed that are better suited for transfer learning.



Development of New Datasets

New datasets are being created that are specifically designed for transfer learning.



Increased Research

There is a growing body of research on transfer learning, leading to new insights and techniques.

THANK YOU